# Iman Jalilvand, Ph.D., EIT

Applied ML Scientist and Simulation Engineer | Reinforcement Learning, XR and Human-Centered Design
Edmonton, Alberta | **Permanent Resident of Canada** | Email | LinkedIn | GitHub | Google Scholar | Portfolio

## Professional summary

Applied ML scientist and simulation engineer with a **Ph.D. in machine learning-based control** and 6+ years taking AI from research to working prototypes with industry partners. Author of **MATRL, a multi-agent transfer reinforcement learning framework** that cut training time by 32% and energy use by nearly 40%. Builds **LLM coaching grounded in validated simulation data** and mixed reality tools in Unity, and proves their value through **user studies and rigorous benchmarking**. Has led six graduate researchers across four industry-supported projects and designed eight university courses.

## Technical skills

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| **Applied AI and RL** | Deep and multi-agent RL (PPO, MAPPO, DQN), transfer learning, reward shaping, domain randomization, Bayesian belief networks, classification and recommendation |
| **LLMs and agents** | LLM application development, grounding in validated system data, prompt design, tool-using and multi-agent workflows, evaluation against ground truth |
| **Programming** | Python (expert), PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, SQL, C#, MATLAB, Git, CI/CD; Claude Code, Cursor, GitHub Copilot |
| **XR and real-time 3D** | Unity (6 years), MRTK, Unity ML-Agents, Microsoft HoloLens 2, Meta Quest, digital twins, Sketchfab |
| **Design and UX** | User-centered design, sketching and wireframing, information architecture, prototyping (Figma, Adobe XD), usability testing (SUS, UMUX) |
| **Research and evaluation** | Experimental design, surveys (Qualtrics), Likert scales, ANOVA, post-hoc tests, correlation, multicriteria decision analysis (TOPSIS), benchmarking and ablation |
| **Engineering** | SolidWorks, AutoCAD, CATIA, ANSYS FEA, COMSOL, heat transfer and process modelling |

## Professional experience

### Research Associate (Postdoctoral Fellow)
Dec 2024 to Present
Materials and Manufacturing Research Institute (MMRI), University of British Columbia

- Developed MATRL, a multi-agent transfer reinforcement learning framework combining Multi-Agent PPO with transfer learning, cutting training time by 32% and energy use by nearly 40% against the multi-agent baseline (Engineering Applications of AI, 2026).
- Designed the FCNN-MAPPO architecture, replacing recurrent policies with state augmentation to lower inference latency and parameter count while improving scalability.
- Built an LLM coaching system that grounds feedback in validated telemetry from a physics-based simulation, delivered in mixed reality; all six study participants improved, cutting zone error by 74.9% on average. US provisional patent application filed (2026).
- Designed data-driven XR dashboards for operator training and decision support, from paper sketches to HoloLens 2, reducing trainees' spatial error by 74% across repeated runs (ACM CHI 2026).
- Built Bayesian belief network models linking process parameters to energy outcomes for explainable, probabilistic decision support (IEOM Europe, 2025).
- Lead and mentor six graduate researchers (three Ph.D., three M.Sc.) across four industry-supported projects, including a feasibility study for the HAZI Foundation in the Basque Country, Spain.
- Contributed to $420K+ in research funding, including a $300K NSERC Alliance grant; grant writer for a $125K NSERC Idea to Innovation proposal.

### Co-Founder and Chief Technology Officer
Present
SustainPro Solutions Inc.

- Lead AI and XR product development, turning life cycle and process data into decision-ready insight for manufacturers.

### Research Consultant (part-time)
Jan 2026 to Present
FEED Clean Tech Inc.

- Advise on data collection, evaluation planning and performance reporting for circular-materials and thermoforming pilots.

### Founder
Sep 2024 to Sep 2025
InnoXR, UBC Venture Founder Program

- Ran 50+ customer discovery interviews across manufacturing, construction and technology, turning operational pain points into prototype requirements; secured support from Kohler, Dassault Systèmes and Cognitive3D.

**AI and XR Developer** Jan 2024 to Jul 2025
Advanced BioCarbon 3D (ABC3D)

- Built Python classification and recommendation models that rank sustainable material candidates for engineering decisions.
- Added real-time 3D visualization that made model outputs interpretable to non-specialists, raising usability scores by 24%.

**Lecturer** 2023 to 2026
University of British Columbia (Computer Science, Engineering, Management); IGK College

- Designed and delivered the first offering of Game Development with Unity (COSC 416) for 152 students, with a GitHub organization of reusable C# modules and a two-week game jam; rated 4.8/5 by students and 6.4/7 by peer review.
- Taught User-Centered Design in Immersive Technologies (IMTC 506, two years, rated 4.84/5) with 10 industry partners, Multicriteria Optimization and Design of Experiments (62 students), Emerging Technologies in Business, and three UI/UX courses.

**Graduate Researcher** Sep 2020 to Dec 2024
Composites Research Network (CRN), University of British Columbia

- Developed a PPO-based deep RL framework for real-time heating control, benchmarked against DQN across 30+ scenarios, reaching 37% faster convergence with lower energy use.
- Built Unity mixed and virtual reality apps coupled to real-time heat simulation for HoloLens 2 and Meta Quest, and designed a 26-participant user study ranked with five multicriteria methods (Journal of Manufacturing Systems, 2024).
- Completed Mitacs internships with lululemon (statistical analysis of fabric test data) and Advanced BioCarbon 3D; Head TA coordinating 12 TAs in CAD/CAM for 500+ students.

**Design Engineer** Mar 2018 to Aug 2019
MD Co., Tehran

- Produced parametric 3D models, assemblies and drawings in SolidWorks and AutoCAD, and verified designs with finite element analysis.

## Education

**Ph.D., Mechanical Engineering (Machine Learning-Based Control)** 2020 to 2025
University of British Columbia. GPA 4.33/4.33. Thesis: Extended reality-based training and adaptive machine learning-based optimization for thermoforming. NSERC CREATE Immersive Technologies (CITech) certificate.

**M.Sc., Mechatronics** 2015 to 2018
Amirkabir University of Technology, Tehran. GPA 4.00/4.33. Structural health monitoring of buried pipelines with piezoelectric sensors.

**B.Sc., Mechanical Engineering** 2011 to 2015
Science and Research Azad University, Tehran. GPA 3.92/4.33. Engineer-in-Training (EIT).

## Selected publications and patent

- Jalilvand, I., et al. (2026). Integrating transfer learning into multi-agent reinforcement learning for energy-efficient adaptive control of heating systems in thermoforming. *Engineering Applications of Artificial Intelligence.* doi:10.1016/j.engappai.2026.114780
- Jalilvand, I., Gopaluni, B., and Milani, A. S. (2026). Data-driven extended reality dashboards: integrating operator training and decision support in manufacturing. *CHI EA '26, Barcelona.* doi:10.1145/3772363.3798455
- Jalilvand, I., and Milani, A. S. (2026). Integrating simulation-based LLM coaching for adaptive mixed-reality manufacturing training. *HCII 2026 Posters, Springer.* doi:10.1007/978-3-032-30816-0_19
- Jalilvand, I., Jang, J., Gopaluni, B., and Milani, A. S. (2024). VR/MR systems integrated with heat transfer simulation for training of thermoforming: a multicriteria decision-making user study. *Journal of Manufacturing Systems, 72.* doi:10.1016/j.jmsy.2023.11.007
- Jalilvand, I., et al. (2025). Optimizing energy-intensive heating processes in manufacturing using Bayesian belief networks. *IEOM Europe, Paris.* doi:10.46254/EU08.20250283
- Jalilvand, I., Milani, A. S., and Gopaluni, B. US Provisional Patent Application 64/093,684 (2026): physics-informed immersive analytics and edge-interface system with adaptive coaching for manufacturing operator decision support.

## Recognition and leadership

- McCall MacBain Postdoctoral Leadership Program, one of 32 national scholars (2025); UBC Dean's Award for Knowledge Translation and Entrepreneurship (2024); Silver Award, HCI International Student Design Competition (2023).
- Co-editor, Elsevier book *Beyond Immersion: Extended Reality Design for Human-Centric Intelligent Systems* (forthcoming); reviewer for ACM CHI 2026 and Robotics and Computer-Integrated Manufacturing; invited speaker, AI Forum, Canada-Korea Conference (2026).